Isaac Soledad Martínez

Financial data platform engineering · Python, React, PostgreSQL · Financial economics

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PROFILE

Financial economist and developer with ten years at a financial advisory firm, having moved from data entry to owning the firm's information. Responsible for the information behind more than 1,800 contracts and close to 2,000 investment accounts held with 30 domestic and international custodians. Designed and built NSC Portal, the internal platform that now supports the firm's operations, information and analytics: 94,152 lines across a Python/FastAPI backend and a React frontend on PostgreSQL, 5 functional applications, 422 endpoints, 77 tables, 47 granular permissions and 203 automated tests. The platform replaced three desktop applications that had no authentication or authorization model, and brought in-house capabilities that were previously licensed. Combines systems engineering with command of the financial content itself —multi-custodian reconciliation, performance measurement, wealth attribution, portfolio analytics and fixed income— backed by the CFA Program, where he is a Level I candidate.

TECHNICAL SKILLS

Backend: Python 3.11, FastAPI, uvicorn, Pydantic, psycopg2; hand-written SQL with no ORM; bcrypt and JWT authentication; authorization declared per endpoint; explicit handling of concurrency between the event loop and the threadpool; job queue persisted in the database.

Frontend: React 18, Vite, react-router, Tailwind; recharts, d3 and lightweight-charts for visualization; bpmn-js for process modelling; File System Access API; custom synchronization and polling hooks.

Data: PostgreSQL: schema design of 77 tables with 71 indexes, migrations from code, monthly aggregation queries with weighting and currency conversion, cache persisted in the database. Multi-source ingestion and reconciliation pipelines (email, SFTP, custodian portals).

Document processing: pdfplumber, PyMuPDF, pikepdf, ocrmypdf, pytesseract, reportlab, python-docx, python-pptx, openpyxl, pyHanko; programmatic manipulation of OOXML.

Quantitative finance: Performance measurement: money-weighted return (Modified Dietz) against time-weighted return, separation of external flows from market effect, attribution by advisor. Portfolio analytics: Sharpe, Sortino, maximum drawdown, Calmar, beta and alpha against a benchmark, VaR and CVaR, risk contribution. Derivatives: implied volatility and greeks with Black-Scholes-Merton. Fixed income: curves, spreads over benchmark and term to maturity on the VALMER price vector.

Data science: pandas, NumPy, scikit-learn, scipy, matplotlib; supervised classification (QDA, LDA, Naive Bayes); clustering (KMeans, DBSCAN, HDBSCAN, agglomerative); dimensionality reduction (PCA, UMAP, t-SNE) and autoencoders; statistical inference implemented from first principles and bootstrap resampling.

Quality and engineering: design for testability through pure functions; mutation testing to verify that tests assert what they claim; code auditing through abstract syntax tree (AST) analysis; catalogue of findings classified by severity and dimension, with a remediation plan ordered by risk-to-cost ratio.

Office and low-code: advanced Excel (Power Query, pivot tables, VBA), Power Apps; automation of document workflows over Office tools.

Information governance: policy catalogue, regulatory frameworks, RACI matrices, BIA analysis, risk maps, document version control; project management.

PROFESSIONAL EXPERIENCE

Information Manager | NSC Asesores — Mexico City
July 2019 – Present

NSC Portal — internal platform for operations, information and analytics

- Cut the area's operational process times by up to 60% by replacing manual review in Excel and repetitive data entry with automated extraction, validation and reporting flows.
- Designed and built the entire platform: Python/FastAPI backend (160 production files, 422 endpoints), React/Vite frontend (101 files, 74 routes) and a 77-table PostgreSQL database, under a single session and a catalogue of 47 granular permissions by application and section.
- Migrated three desktop Streamlit applications (41,507 lines, three independent processes, no authentication or authorization) to a single HTTP service, adding JWT sessions, an audit log, a job queue persisted in the database that survives restarts, and deployment independent of the user's machine.
- Built the ingestion and reconciliation pipeline for balances and transactions from 30 domestic and international custodians: 15 PDF statement readers behind a common base class, plus automated daily collection from email, SFTP and portals, with previous-business-day resolution against a holiday calendar for three countries and a dry-run mode before touching production files.
- Implemented the wealth business intelligence engine: monthly SQL aggregation over balances, positions and transactions; wealth attribution weighted by advisor share; and a net new money calculation that separates real flow from market effect, internal transfers and initial deposits, with exceptions, reclassifications and a correction log.
- Developed the internal markets terminal: a provider router with graceful degradation and circuit breakers, a shared cache in PostgreSQL, daily ingestion of the VALMER price vector (roughly 35,000 instruments by 76 columns) and portfolio and derivatives analytics computed locally.
- Replaced paid licenses and services with in-house computation: a BPMN modeller with an execution engine instead of the commercial modelling tool; a full PDF editing, OCR and digital signature suite instead of a licensed desktop editor; and ETF look-through from N-PORT regulatory filings instead of a paid data provider.
- Implemented a per-currency return calculation that separates positions and flows into peso and dollar buckets, with invested capital weighted by each flow's days of exposure —a Modified Dietz approximation to money-weighted return — replacing a figure the advisor had been computing by hand.
- Automated document output: wealth report in PDF, executive deck in PPTX on the corporate template, letters generated from scratch in DOCX, and exports to Excel and email, including an audit trail of the calculation.
- Established the quality practice: 203 automated tests focused on the arithmetic of money and on the session —the silent error, not the visible one— verified through mutation testing; plus an in-house audit of 99 findings citing file and line, classified by severity and dimension, with a remediation plan ordered by risk-to-cost ratio.

Operations, control and information governance

- Single point of consolidation for the institution's securities information: transactions, positions, prices, account statements and withholding certificates, and its delivery to accounting, control, sales and the advisor network.
- Designed the area's information governance framework: a catalogue of 12 policies, role definitions, a RACI matrix and a clear boundary between what the area governs directly and what requires cross-area agreement.
- Led the update of the Portal Access and Use Policy (v5.1 → v6.0), resolving ten review observations with substantive decisions on scope, the definition of confidential information, prohibitions and the reassignment of monitoring responsibility.

IT Support Assistant | NSC Asesores — Mexico City
July 2017 – May 2019

- User support and administration of internal equipment and systems; first automations over the area's operational databases.

Social service — Anti-Money Laundering | Sociedad Hipotecaria Federal

2023

- Cleaning and consolidation of databases for client risk analysis, and updating of the datasets used in financial risk assessment models.

EDUCATION

M.Sc. in Data Science and Information | INFOTEC

July 2025 – July 2027 (in progress)

Bachelor's in Economics and Finance | Universidad del Valle de México

Graduated 2023 · Professional license (cédula) 15462554

Specialization in Project Management | Tecnológico de Monterrey

August 2024

Specialization in Data Science | Universidad de los Andes, Colombia

November 2023

Diploma: Computer Systems Technician | Fundación Carlos Slim

March 2020

CERTIFICATIONS AND LANGUAGES

- CFA Program — Level I Candidate, exam scheduled for November 2026.
- Google Data Analytics Professional Certificate — SQL, BigQuery, R and visualization.
- Certificate in Alternative Investments — CAIA Association and iCapital (2026).
- Python Data Associate (2026) · SQL Associate (2026) · Data Literacy Certificate.
- Microeconomics Principles · Alts Edge, Alternative Investments · Innovation, Transformation and Leadership.
- English: TOEIC 920, professional working proficiency. Native Spanish.

ACADEMIC AND PERSONAL PROJECTS

- Comparison of dimensionality reduction methods on California Housing: PCA, UMAP, t-SNE and an autoencoder, analysing the structure each method recovers.
- Unsupervised segmentation evaluated with silhouette coefficient and intra/inter-cluster metrics, comparing KMeans, DBSCAN, HDBSCAN and agglomerative clustering.
- Mobile training-tracking application in Flutter on Firebase: layered architecture, versioned data model, record-level privacy controls and autoregulated progression rules.